

Large Language Models on HPC

Day 3

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Now we will walk through building a container for Llama.cpp
and interacting with an LLM together

Open **Day3_LLM.md**

One-Liners

Change out the text string in: -p "Hello from the HPC"

1. List five good study habits.
2. Briefly explain some methods researchers use to analyze large datasets.
3. Generate six short icebreaker questions for an introductory sociology course.
4. List 3 strategies for breaking down complex concepts.
5. State one key reason why peer-review is essential in academic research.

Batch script

```
#!/bin/bash
#SBATCH --job-name=llama_infer
#SBATCH --output=output_llama_infer.%A
#SBATCH --error=error_llama_infer.%A
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --time=00:30:00
#SBATCH -p gpu
#SBATCH -q gpu
#SBATCH --gres=gpu:1

set -euo pipefail
module load apptainer

# Path to your container
CONTAINER="$HOME/llm/llama_cuda.sif"

time srun apptainer exec --nv -B /grps2:/grps2 "$CONTAINER" \
  llama-cli \
    -m /grps2/ualib/llm/phi-4-Q6_K.gguf \
    -n 512 \
    --threads "$SLURM_CPUS_PER_TASK" \
    --ctx-size 0 \
    -ngl 999 \
    --single-turn \
```

A mobile app suddenly hits API rate limits after a new feature rollout.

- Identify likely coding or architectural causes
- Explain how you would instrument the system to diagnose the issue
- Suggest two mitigation strategies
- Present your answer in a table with "Cause" and "Fix"

Adjusting the prompt in a batch script

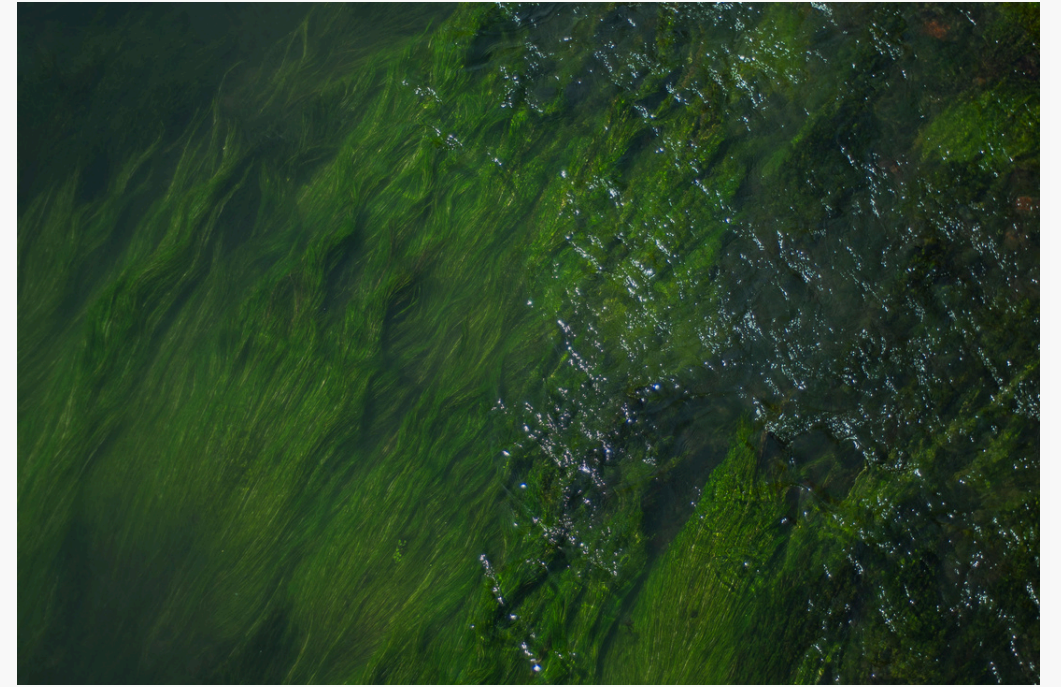
We can directly change the prompt for the model by adjusting the argument `-p` or --prompt`.`

Let's ask the model to give us some travel tips:

Plan an affordable weekend trip within driving distance from Tuscaloosa, Alabama



Submitting a text file prompt



"Describe how an AI system could assist in detecting early signs of harmful algal blooms.

Identify the types of data required and potential sources of error.

Explain how model outputs should be validated by domain experts.

Provide a short analysis of risks and limitations. Present your answer in a structured list."

Changing Parameters

By changing model parameters, we can optimize:

- Quality of responses
- Creativity
- Speed
- Stability
- Adjust to hardware

Run this command:

```
nano text_inf.bat
```

Adjusting Context Size

The context size , `--ctx-size`, is how much the model can remember at once.

- Prompts
- Chat template text
- Conversation history

A “too small” context window will cause the model to

- Forget instructions
- Lose details
- Contradict itself

Adjusting Context Size

In the batch file, we'll expand the context size:

```
-m $HOME/llm/models/phi-4-Q6_K.gguf \  
-n 512 \  
--threads "$SLURM_CPUS_PER_TASK" \  
--ctx-size 3000 \  
-ngl 999 \  
--single-turn \  

```

Maximum Number of Tokens

-n determines the output length by setting the maximum number of tokens to generate.

You should change **-n** in the following ways:

- Outputs are getting cut off --> increase **-n**
- The model rambles --> decrease **-n**

Maximum Number of Tokens

In the batch file, we'll increase the max number of tokens:

```
-m $HOME/llm/models/phi-4-Q6_K.gguf \  
-n 800 \  
--threads "$SLURM_CPUS_PER_TASK" \  
--ctx-size 3000 \  
-ngl 999 \  
--single-turn \  

```

Comparing models

phi-4 vs. gpt-oss 20b vs. gpt-oss 120b

1. How is the weather today?
2. If all birds can fly, and a penguin is a bird, can a penguin fly?
3. Please provide a brief summary of the primary purpose and key outcomes of the Apollo 11 mission.